

Assignment Report

Question 1: SVM Classification on Pulsar Dataset

1.Data Preparation

The dataset contains 8 numerical features derived from the pulsar signal. Missing values were handled using mean imputation. Data was split into train and validation sets.

2. Model Building

Support Vector Machine (SVM) models were trained using the following kernels:

- Linear
- Polynomial
- Sigmoid

3. Evaluation

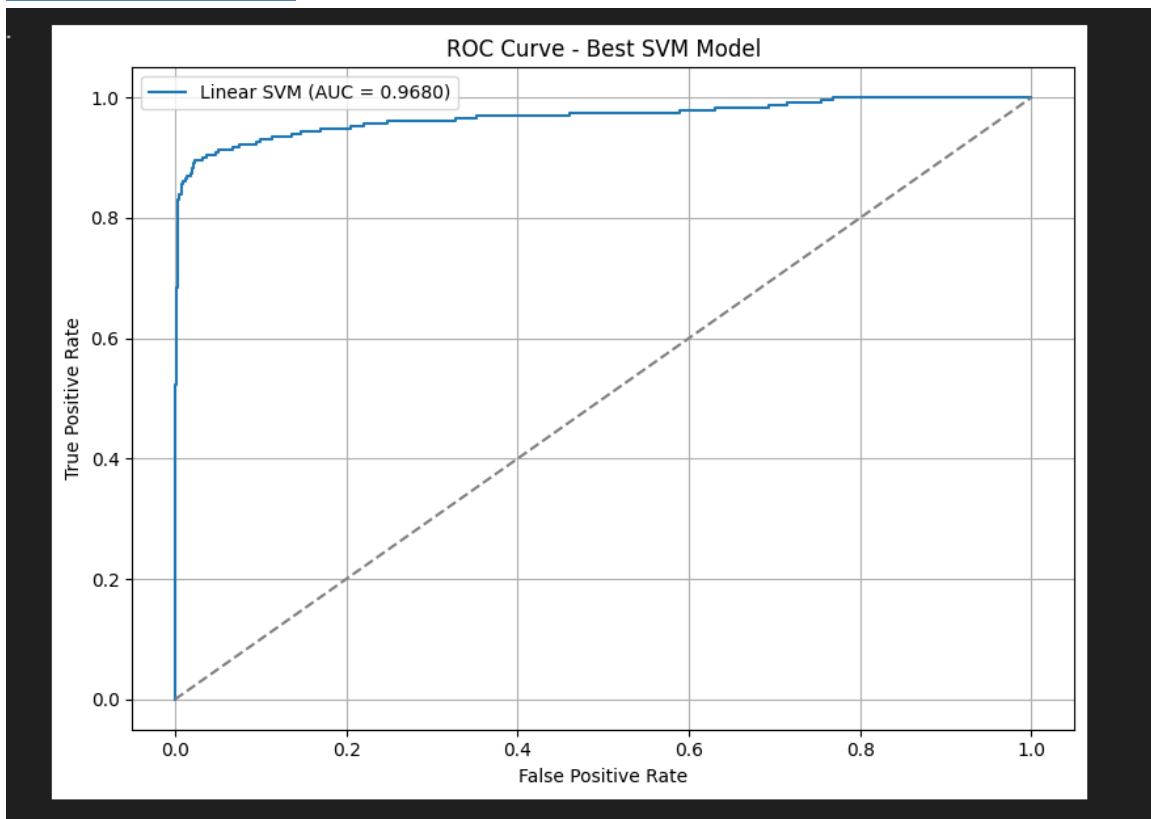
The models were evaluated using confusion matrix, classification report, ROC-AUC score, and Stratified K-Fold Cross Validation.

Linear SVM performed best with an accuracy of 97.9% and ROC-AUC of 0.968.

4. Hyperparameter Tuning

GridSearchCV was used to optimize hyperparameter C. Best value found was C=10.

5.ROC CURVE



6. Test Set Prediction

The optimized linear SVM model was used to make predictions on the test set. Results were saved to 'svm_pulsar_predictions.csv'.

7. Conclusion

Linear SVM was most effective for pulsar classification. It outperformed other kernels in accuracy and ROC-AUC. The sigmoid kernel underperformed. Model tuning improved accuracy further.

Question 2: Bagging and Gradient Boosting on MNIST & Fashion MNIST

1. Bagging Implementation

Implemented manual Bagging using DecisionTreeClassifier as base learner. Bootstrap sampling was used to train 10 estimators. Final predictions were obtained using majority voting.

2. Gradient Boosting Implementation

Implemented custom Gradient Boosting using DecisionTreeRegressor with max_depth=1 (stumps). Each tree learned residuals of previous trees, and predictions were aggregated and converted into class labels.

3. Evaluation

MNIST - Bagging:

Accuracy: 87.7%

MNIST - Gradient Boosting:

Accuracy: 49.3%

MAE: 1.762

R2 Score: -0.0086

Fashion MNIST - Bagging:

Accuracy: 82.9%

Fashion MNIST - Gradient Boosting:

Accuracy: 59.1%

MAE: 1.255

R2 Score: 0.3902

4. Conclusion

Bagging outperformed custom boosting on both datasets. Gradient boosting was affected by limited depth and small estimator count, but still showed decent performance. Bagging showed strong robustness and accuracy.

```
=== MNIST - Bagging ===
Accuracy: 0.8975
Confusion Matrix:
[[188  0  1  0  0  0  0  3  3  2]
 [  0 217  1  1  3  1  0  1  0  1]
 [  6  3 176  0  3  2  2  1  4  2]
 [  1  1  5 186  1  1  1  1  5  2]
 [  4  1  1  0 171  0  5  2  5  6]
 [  4  1  1 10  3 152  3  2  3  2]
 [  4  2  5  1  0  1 178  0  6  0]
 [  1  2  4  3  3  0  0 187  1  8]
 [  2  3  3  3  3  5  3  1 166  6]
 [  1  1  3  3  5  3  2  5  1 174]]
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=== MNIST - Gradient Boosting ===
Accuracy: 0.4925
Confusion Matrix:
[[125  28   1   2  16   0   7   0  12   6]
 [  0 184   0   5  29   0   1   0   4   2]
 [ 13  28  56  10  15   0  38   0  37   2]
 [  3  36   7 112  23   0   4   5   9   5]
 [ 12   1   0   3 133   9   1   2  23  11]
 [  5  37   8  28  16  57   9   3  11   7]
 [ 34  11   4   1  41   1  88   0  15   2]
 [  8   5   0   4  17   0   0 111  16  48]
 [ 11  57   2  18  27  16   0   0  56   8]
 [  5   8   1  20   3   0   0  59  39  63]]
MAE: 1.762
R2 Score: -0.008649604866162175

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=== Fashion MNIST - Bagging ===
Accuracy: 0.8125
Confusion Matrix:
[[163   0   2  10   3   2  19   0   1   0]
 [  1 190   3   4   0   0   2   0   0   0]
 [  4   0 157   3  22   1  13   0   0   0]
 [  4   3   2 178   7   0   5   0   1   0]
 [  2   0  31  15 134   0  15   0   3   0]
 [  0   0   0   1   0 181   0  10   3   5]
 [ 30   1  34   6  29   0  95   0   5   0]
 [  0   0   0   0   0  23   0 169   0   8]
 [  1   0   1   2   3   3   4   1 185   0]
 [  0   0   0   1   0   6   0  19   1 173]]

```

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=== Fashion MNIST - Gradient Boosting ===
Accuracy: 0.5910
Confusion Matrix:
[[131   3   2  17  11   5  24   0   6   1]
 [  1 163   0  27   2   0   5   0   0   2]
 [ 57   0  96  15   9   1   8   9   4   1]
 [  2   2   0 133  42   3  12   0   0   6]
 [  7   0  82  17  71   1   8   9   3   2]
 [  0 20   0   2   0 104   0  27   6  41]
 [ 41   1  50  11  39   3  44   2   8   1]
 [  0 14   0   2   1   7   0 131   3  42]
 [  3   1  17   4  25   4   1  10 130   5]
 [  0   0   0   1   0   0   0   8  12 179]]
MAE: 1.2555
R2 Score: 0.3902424242424243
```